

Catalysis Science & Technology

www.rsc.org/catalysis

Volume 1 | Number 7 | October 2011 | Pages 1085–1272

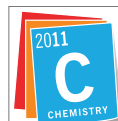


ISSN 2044-4753

RSC Publishing

COVER ARTICLE

Ishihara *et al.*
Catalytic enantioselective alkyl and aryl addition to aldehydes and ketones with organozinc reagents



International Year of
CHEMISTRY
2011



2044-4753(2011)1:7;1-A

Cite this: *Catal. Sci. Technol.*, 2011, **1**, 1149–1158

www.rsc.org/catalysis

PAPER

Catalytic enantioselective alkyl and aryl addition to aldehydes and ketones with organozinc reagents derived from alkyl Grignard reagents or arylboronic acids†

Manabu Hatano,^a Riku Gouzu,^b Tomokazu Mizuno,^a Hitoshi Abe,^b
Toshihide Yamada^b and Kazuaki Ishihara^{*ac}

Received 28th March 2011, Accepted 2nd June 2011

DOI: 10.1039/c1cy00108f

A highly practical, catalytic enantioselective alkyl and aryl addition to aldehydes and ketones with organozinc reagents, which were prepared *in situ* from commercially available Grignard reagents or arylboronic acids, was developed. A chiral phosphoramidate ligand was essential for promoting the addition reactions in high yields with high enantioselectivities.

1. Introduction

Catalytic enantioselective organozinc addition to aldehydes and ketones is an important synthetic approach that gives optically active secondary (2°) and tertiary (3°) alcohols, respectively, *via* carbon–carbon bond-formation under mild conditions.¹ However, in sharp contrast to many examples of the di(1°-alkyl)zinc addition to aldehydes, only a few examples of the di(2°-alkyl)zinc addition to aldehydes and ketones have been reported.² Moreover, the addition of arylzinc reagents to less-reactive ketones rather than aldehydes has not been well-established. These difficulties are due to steric constraints between substrates and reagents, particularly when ketones are used. In addition, the fact that di(2°-alkyl)zinc and arylzinc reagents are not available is another serious problem. Therefore, particularly for industrial use, inexpensive, easily handled, and low-toxicity alkyl and aryl sources for organozinc reagents are needed in large quantities. In this regard, Grignard reagents and boronic acids are highly attractive not only for academic reasons but also for industry, since a variety of these reactants are commercially available or readily prepared from the corresponding halides. Recently, some useful and practical methods for preparing pure organozinc reagents *in situ* from these more convenient organometallic reagents have been developed.^{3–6} In this context, we report here a catalytic enantioselective alkyl and aryl addition to aldehydes and ketones with a variety of organozinc reagents,

which are prepared *in situ* from alkyl Grignard reagents or arylboronic acids as convenient alkyl or aryl sources.

2. Results and discussion

2.1. Grignard reagent-derived di(2°-alkyl)zinc addition to aldehydes and ketones

Traditional Grignard reagents are some of the most useful and safe reagents for alkylating carbonyl compounds in academic and industrial research on process chemistry. Today, more than 200 inexpensive Grignard reagents can be purchased. In particular, the transmetalation of alkyl groups in Grignard reagents to other less-reactive metal complexes such as Zn(II) should be attractive for catalytic enantioselective alkylation to carbonyl compounds because the reaction would proceed only in the presence of the catalysts. Although convenient traditional methods have been reported for the preparation of diorganozinc reagents *in situ*, they are usually accompanied by the generation of inorganic salts, which sometimes disturb subsequent asymmetric catalysis. In contrast, Côté and Charette recently developed a highly useful method for the preparation of salt-free di(1°-alkyl)zinc reagents from ZnCl₂, NaOMe, and Grignard reagents.⁶ This method is highly practical for small scale application in both laboratory and industry since centrifugation is needed to separate the salt-free dialkylzinc reagents from the resultant magnesium salts. By taking advantage of Charette's procedure, we here prepared salt-free di(2°-alkyl)zinc reagents for the catalytic enantioselective 2°-alkylation of aldehydes. First, the isopropylation of benzaldehyde (**2a**) was investigated with chiral phosphoramidate ligand **1b** (10 mol%) (Table 1). By following Charette's typical procedure for the preparation of di(1°-alkyl)zinc,⁶ we prepared a 0.44 M solution of *i*-Pr₂Zn in Et₂O, but nearly racemic **3a** was obtained in 90% yield under Et₂O conditions (entry 1). We assumed that a small amount of the remaining

^a Graduate School of Engineering, Nagoya University, Furo-cho, Chikusa, Nagoya, 464-8603, Japan.
E-mail: ishikawa@cc.nagoya-u.ac.jp; Fax: +81-52-789-3331;
Tel: +81-52-789-3222

^b Sekisui Medical Co., Ltd, 4-115, Matsuo, Hachimantai, Iwate 028-7305, Japan

^c Japan Science and Technology Agency (JST), CREST, Furo-cho, Chikusa, Nagoya, 464-8603, Japan

† Electronic supplementary information (ESI) available: Experimental details and product characterization. See DOI: 10.1039/c1cy00108f

Table 1 Optimization of the isopropylation of benzaldehyde^a

[ZnCl₂ + NaOMe + *i*-PrMgCl in Et₂O]

rt for 2 h, then centrifugation
(-NaOMe, -NaCl, -Zn(OMe)₂, -MeOMgCl)
concentration (-Et₂O)

PhCHO (2a) + *i*-Pr₂Zn (salt free, 3 equiv) $\xrightarrow[1b (10 \text{ mol\%})]{\text{solvent free, rt, 2 h}}$ PhCH(OH)*i*-Pr (3a)

1a **1b** (-)-MIB

Entry	Molar ratio of ZnCl ₂ : NaOMe : <i>i</i> -PrMgCl	Ligand	Yield [%] of 3a	ee [%] of 3a
1 ^b	1 : 2 : 1.9	1b	90	3
2 ^{b,c}	1 : 2 : 1.6	1b	73	91
3	1 : 2 : 1.6	1b	93	91
4	1 : 2.5 : 1.6	1b	94	94
5	1 : 2.5 : 1.6	1a	93	93
6	1 : 2.5 : 1.6	(-)-MIB	85	59
7 ^d	1 : 2.5 : 1.6	(-)-MIB	85	94

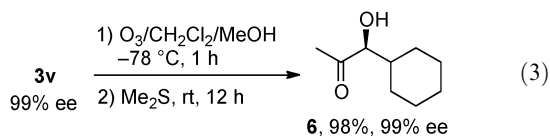
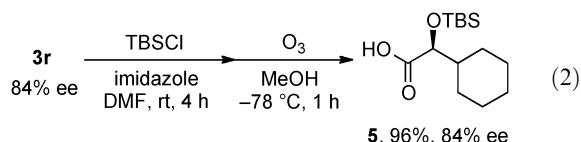
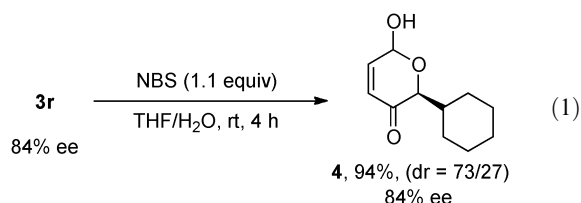
^a Unless otherwise noted, salt-free *i*-Pr₂Zn was used under solvent-free conditions. ^b *i*-Pr₂Zn (0.44 M, in Et₂O) was used. ^c BnOH was obtained in 26% yield. ^d Toluene was used as a solvent.

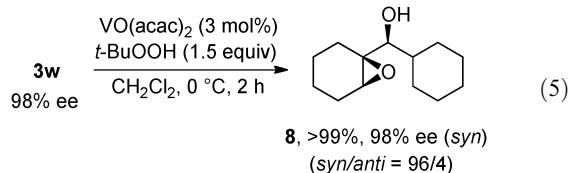
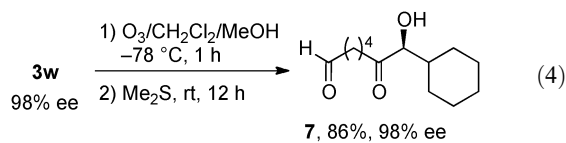
Grignard reagent might trigger the racemic pathway, since a highly active zinc(II) ate complex for alkylation, namely [*i*-Pr₃Zn][MgCl]⁺, would be generated *in situ*.⁷ As expected, a conservative molar ratio of 1/2/1.6 of ZnCl₂/NaOMe/*i*-PrMgCl was effective, and **3a** was obtained in 73% yield with 91% ee (entry 2). However, BnOH was also obtained as an undesired reduction byproduct in 26% yield (entry 2). Therefore, we examined the reaction under solvent-free conditions⁸ by using salt-free liquid *i*-Pr₂Zn (bp 134 °C),⁹ which was prepared by the same method, followed by the removal of Et₂O *in vacuo*. As a result, **3a** was obtained in an improved yield (93%) with 91% ee, along with a trace amount of BnOH (<2%) (entry 3). Moreover, with the use of 2.5 equiv. of NaOMe, **3a** was obtained in 94% yield with 94% ee (entry 4). Both less-bulky **1b** (10 mol%) and more-bulky **1a**¹⁰ were effective in this reaction (entries 4 and 5). Côté and Charette used (2*S*)-(-)-3-*exo*-(*N*-morpholino)isoborneol [(-)-MIB], which is a representative chiral ligand,¹¹ in their asymmetric 1°-alkylation to aldehydes.⁶ However, (-)-MIB was less effective than chiral ligand **1** in the isopropylation of **2a** under solvent-free or conventional toluene conditions (entries 6 and 7).

Encouraged by the efficient isopropylation of **2a**, we next examined the catalytic enantioselective 2°-alkyl addition to various aldehydes under solvent-free conditions with salt-free di(2°-alkyl)zinc reagents derived from Grignard reagents (Table 2). The isopropylation of aromatic aldehydes (entries 1 and 2), heteroaromatic aldehydes (entries 3 and 4), and cycloaliphatic aldehydes (entries 5 and 6) proceeded, and the

corresponding products were obtained in high yields with high enantioselectivities (80–>99% ee). The isopropylation of tiglic aldehyde as an α,β-unsaturated aldehyde provided only a 1,2-adduct (**3h**) with high enantioselectivity (97% ee) (entry 7). Catalytic enantioselective *sec*-butylation also proceeded for the first time, and the desired products were obtained with high enantioselectivities (94–98% ee) (entries 8 and 9). Next, as an unprecedented cyclic 2°-alkyl addition to aldehydes, the addition of (*c*-C₅H₉)₂Zn and (*c*-C₆H₁₁)₂Zn was explored. Since (*c*-C₆H₁₁)₂Zn is viscous at room temperature, toluene (2.5 M) was used as a solvent. Fortunately, the desired products were obtained with high enantioselectivities from a variety of aromatic aldehydes (entries 10, 13–16), heteroaromatic aldehydes (entries 11 and 17), cycloaliphatic aldehydes (entries 12 and 18), α-branched or β-branched aliphatic aldehydes (entries 19 and 20), and α,β-unsaturated aldehydes (entries 21 and 22). Ligand **1a** generally showed high reactivity,¹⁰ while less-bulky **1b** was often less effective than bulky **1a** due to reduction byproducts, especially when less-reactive aliphatic aldehydes were used (entries 9, 18, and 22). Probably, bulky ligand **1a** would be essential for keeping active monomeric catalysts *in situ* and thus for establishing both high yield and high enantioselectivity.¹⁰ Some of the secondary alcohols (**3**, R'RCHOH) which have two similar cyclic and/or acyclic fragments (R' and R) can hardly be obtained *via* complementary methods such as asymmetric reduction of the corresponding ketones.¹² In this catalysis, optically active novel secondary alcohols with similar R' and R were successfully obtained with high to excellent enantioselectivities.

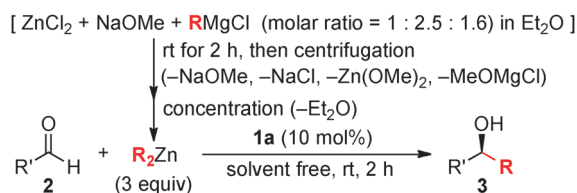
To demonstrate the synthetic utility of this approach, several transformations from the obtained optically active alcohols were examined. The optically active cyclohexyl-(furan-2-yl)methanol (**3r**) was treated with *N*-bromosuccinimide in THF–H₂O at room temperature for 4 h,¹³ and the desired cyclohexyl-substituted 6-hydroxy-2*H*-pyran-3-one **4**¹⁴ was obtained *via* oxidative Achmatowicz rearrangement in 89% yield with a diastereomeric ratio of 73:27 with 84% ee (eqn (1)). Moreover, after TBS-protection of the hydroxy group of **3r**, ozonolysis cleavage of the furan ring provided α-alkoxy carboxylic acid **5**^{15,16} in 96% yield (eqn (2)).

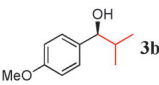
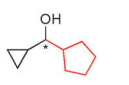
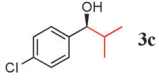
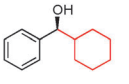
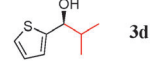
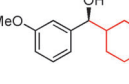
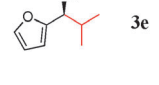
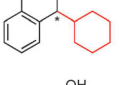
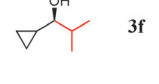
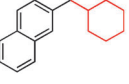
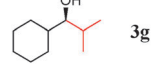
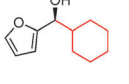
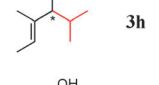
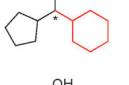
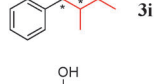
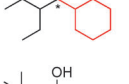
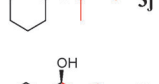
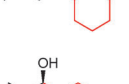
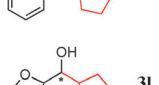
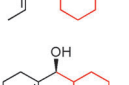
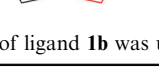
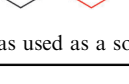




By ozonolysis and subsequent Me_2S treatment,¹⁷ allyl alcohol **3v** was directly converted to α -hydroxy ketone **6**¹⁸ in 98% yield with 99% ee (eqn (3)). This transformation is synthetically important because compound **6** cannot be prepared directly from unstable methylglyoxal by alkylation. Similarly, α -hydroxy ketone **7** bearing a terminal formyl moiety was readily prepared from allyl alcohol **3w** in 86% yield with 98% ee (eqn (4)). For another oxidative transformation, the diastereoselective epoxidation of allyl alcohol **3w** with *tert*-butyl hydroperoxide (TBHP) in the presence of vanadyl acetylacetonate was examined (eqn (5)).¹⁹ Fortunately, *syn*-epoxide **8**,

Table 2 Enantioselective 2°-alkyl addition to aldehydes



Entry	Product (3)	Yield and ee	Entry	Product (3)	Yield and ee
1 ^a	 3b	95%, 96% ee	12 ^b	 3m	90%, 99% ee
2	 3c	90%, 80% ee	13	 3n	78%, 90% ee
3 ^a	 3d	>99%, 95% ee	14 ^{a,c}	 3o	89%, 97% ee
4	 3e	86%, 90% ee	15 ^{a,c}	 3p	96%, 90% ee
5	 3f	>99%, 96% ee	16	 3q	91%, 95% ee
6 ^a	 3g	98%, >99% ee	17 ^c	 3r	94%, 84% ee
7 ^b	 3h	97%, 97% ee	18 ^c	 3s	77% [36%] ^a , >99% ee
8	 3i	76%, (dr 55/45) 94% ee / 95% ee	19 ^{b,c}	 3t	80%, 99% ee
9	 3j	45% [15%] ^a , (dr 59/41) 98% ee / 96% ee	20 ^{b,c}	 3u	75%, 82% ee
10	 3k	52%, >99% ee	21 ^c	 3v	85%, 99% ee
11	 3l	56%, 96% ee	22 ^c	 3w	88% [42%] ^a , 98% ee

^a 10 mol% of ligand **1b** was used in place of **1a**. ^b Temperature was 0 °C. ^c Toluene (2.5 M) was used as a solvent.

Table 3 Enantioselective 2°-alkyl addition to ketones

$$[\text{ZnCl}_2 + \text{NaOMe} + \text{R}^*\text{MgCl} \text{ (molar ratio} = 1 : 2.5 : 1.6) \text{ in Et}_2\text{O}]$$

rt for 2 h, then centrifugation
(–NaOMe, –NaCl, –Zn(OMe)₂, –MeOMgCl)
concentration (–Et₂O)

Entry	Ligand	Product (Ar, R)	Yield and ee of 10	Yield of 11	Recovery of 9
1 ^a	(–)-MIB	10a (4-CF ₃ C ₆ H ₄ , <i>i</i> -Pr)	0%, —	40%	60%
2 ^b	1a	10a (4-CF ₃ C ₆ H ₄ , <i>i</i> -Pr)	38%, >99% ee	<3%	59%
3 ^b	1a	10b (3,5-(CF ₃) ₂ C ₆ H ₃ , <i>i</i> -Pr)	40%, >99% ee	<3%	57%
4 ^c	1a	10c (4-CF ₃ C ₆ H ₄ , <i>c</i> -Hex)	40%, >99% ee	<3%	57%
5 ^c	1a	10d (3,5-(CF ₃) ₂ C ₆ H ₃ , <i>c</i> -Hex)	56%, >99% ee	<3%	41%

^a 20 mol% of (–)-MIB was used. Toluene was used as a solvent. ^b 10 mol% of **1a** was used under solvent-free conditions. ^c 20 mol% of **1a** was used in 2.5 M toluene for **9**.

which is a key intermediate in the synthesis of optically active 1,3-diols with three consecutive chiral carbon centers,²⁰ was obtained in a diastereomeric ratio of 96:4 with 98% ee. Other oxidation methods^{19c} such as *m*-CPBA, Ti(O*i*-Pr)₄/TBHP, Mo(CO)₆/TBHP were less effective than VO(acac)₂/TBHP and gave poor to moderate diastereoselectivities (up to 80/20 dr).

Finally, the 2°-alkylation of ketones in place of aldehydes was examined (Table 3). A preliminary (–)-MIB catalysis in the isopropylation of 4'-(trifluoromethyl)acetophenone (**9a**) did not proceed and undesired aldol byproduct **11a** was obtained along with recovery of the starting ketone **9a**. In sharp contrast, the isopropylation of **9a** and 3',5'-bis(trifluoromethyl)acetophenone (**9b**) proceeded without aldol formation at 0 °C for 24 h in the presence of 10 mol% of ligand **1a**, and the desired tertiary alcohols (**10a** and **10b**) were obtained in moderate yield with >99% ee (entries 2 and 3). Furthermore, the cyclohexylation of **9a** and **9b** in the presence of 20 mol% of ligand **1a** provided the desired products **10c** and **10d** in moderate to good yield with >99% ee (entries 4 and 5). Although further improvements with regard to yield and the use of less-reactive substrates will be necessary, to the best of our knowledge, this is the first example of catalytic asymmetric tertiary alcohol synthesis *via* di(2°-alkyl)zinc addition to ketones.²¹

2.2. A concise synthesis of (S)-(+)-ginnol based on a catalytic enantioselective long *n*-alkyl chain addition to aldehydes

The catalytic enantioselective long *n*-alkyl chain addition to aldehydes has thus far been limited, since the corresponding di(*n*-alkyl)zinc reagents are difficult to prepare.²² By taking advantage of the preparation of solvent-free di(2°-alkyl)zinc reagents as above, we next examined a long-chain-alkylation to aldehydes, such as *n*-heptylation, *n*-octylation, *n*-nonylation, and *n*-decylation.

First, we prepared di(*n*-alkyl)zinc reagents and performed the subsequent addition reaction to aromatic aldehydes such as benzaldehyde and *p*-anisaldehyde under the optimized conditions with the use of chiral ligand **1a** (Table 4). *n*-Heptylation,

Table 4 Enantioselective long *n*-alkyl chain addition to aldehydes

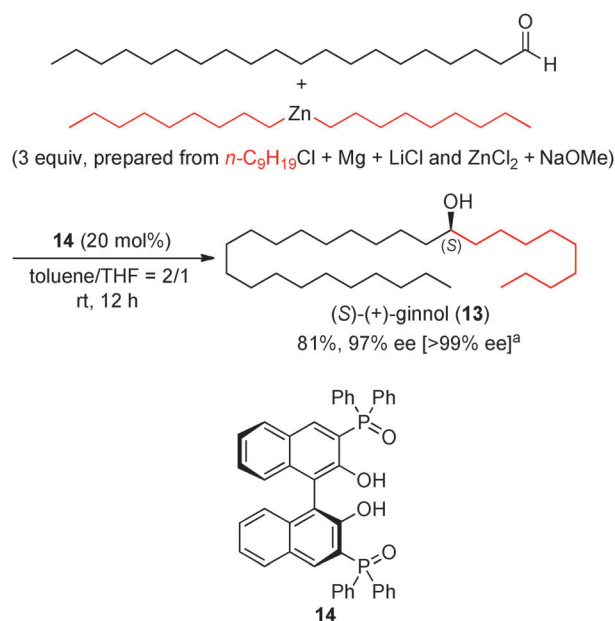
$$[\text{ZnCl}_2 + \text{NaOMe} + \text{R}^*\text{MgCl} \cdot \text{LiCl} \text{ (molar ratio} = 1 : 2.5 : 1.6) \text{ in Et}_2\text{O}]$$

rt for 2 h, then centrifugation
(–NaOMe, –NaCl, –Zn(OMe)₂, –MeOMgCl)
concentration (–Et₂O)

Product, yield, enantioselectivity, and alkyl source	
12a, 98%, 92% ee (<i>n</i>-C₇H₁₅Cl + Mg + LiCl)	12b, >99%, 94% ee (<i>n</i>-C₇H₁₅Cl + Mg + LiCl)
12c, 98%, 96% ee (<i>n</i>-C₈H₁₇Cl + Mg + LiCl)	12d, >99%, 95% ee (<i>n</i>-C₁₀H₂₁Cl + Mg + LiCl)

n-octylation, and *n*-decylation proceeded smoothly, and the corresponding secondary alcohols (**12**) were obtained in almost quantitative yields with 92–96% ee. Remarkably, the use of *in situ*-prepared Grignard reagents (RCl + Mg + LiCl²³) appeared to have no negative effect.

As an application of this approach to an aliphatic aldehyde, we next examined the synthesis of a chiral long-chain aliphatic alcohol, (S)-(+)-ginnol (**13**),²⁴ which is an optically active natural product that has been noted in chiral bilayers of the wax tubes of higher plants. Unfortunately, the reaction of icosanal (*n*-C₁₉H₃₉CHO) with di(*n*-nonyl)zinc [(*n*-C₉H₁₉)₂Zn] in the presence of chiral ligand **1a** (20 mol%) failed, and a complex mixture was obtained. In sharp contrast, our previous chiral BINOL-derived ligand **14** under toluene/THF = 2/1



Scheme 1 Synthesis of (S)-(+)-ginnol (**13**). (^aEnantioselectivity after recrystallization from *n*-hexane.)

conditions²⁵ was effective, and enantio-enriched (S)-(+)-ginnol (**13**) was obtained in one step in 81% yield with 97% ee at room temperature within 12 h (Scheme 1). To the best of our knowledge, this is the first one-step example, and thus this is the simplest asymmetric synthesis of (S)-(+)-ginnol. Fortunately, a single recrystallization in *n*-hexane provided optically pure (S)-(+)-ginnol ($>99\%$ ee).

2.3. Arylzinc addition to ketones with arylboronic acids as an aryl source

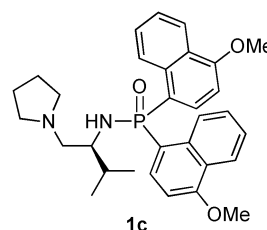
Commercially available diphenylzinc has been widely used in the catalytic enantioselective phenylation of aldehydes and ketones.²⁶ However, diphenylzinc is not inexpensive, and other diarylzinc reagents are not readily commercially available. In this regard, some alternative methods for preparing arylzinc reagents, which are suitable for catalytic enantioselective aryl addition, have recently been developed.²⁷ One of the most attractive methods was reported by Bolm *et al.*, and arylzinc reagents were prepared *in situ* from diethylzinc (7.2 equiv.) and arylboronic acids (2.4 equiv.) as an aryl source.²⁸ Arylboronic acids are generally non-toxic, easily handled, and have recently become commercially available in bulk quantities since these reagents are used in the Suzuki–Miyaura cross-coupling reaction. It has since been shown that triarylboroxines [(ArBO)₃] could be used in place of arylboronic acids to reduce the amount of diethylzinc needed.²⁹ By taking advantage of these developments with arylboronic acids and triarylboroxines, we next examined catalytic enantioselective arylation to ketones. Unlike aldehydes, catalytic enantioselective arylation to ketones with arylzinc reagents has not yet been well-explored due to the extremely low reactivities of ketones.^{30,31}

We first investigated catalytic enantioselective phenylation to 4'-chloroacetophenone (**9c**) with phenylboronic acid (**15a**) (Table 5). In this reaction, triphenylboroxine (**16a**) can be readily prepared *in situ* in toluene under azeotropic reflux

Table 5 Enantioselective phenylation to 4'-chloroacetophenone (**9c**)

Entry	Et ₂ Zn (equiv.)	Concentration ^a /M	Yield (%) of 17a	ee (%) of 17a	Yield (%) of 18
1	3.0	0.17	70	94 (S)	18
2	4.5	0.17	86	94 (S)	11
3	4.5	0.25	79	94 (S)	9
4	4.5	0.14	68	92 (S)	10
5 ^b	4.5	0.17	83	94 (S)	13

^a Concentration was based on **9c** in toluene. ^b Chiral ligand **1c** was used in place of **1a**.



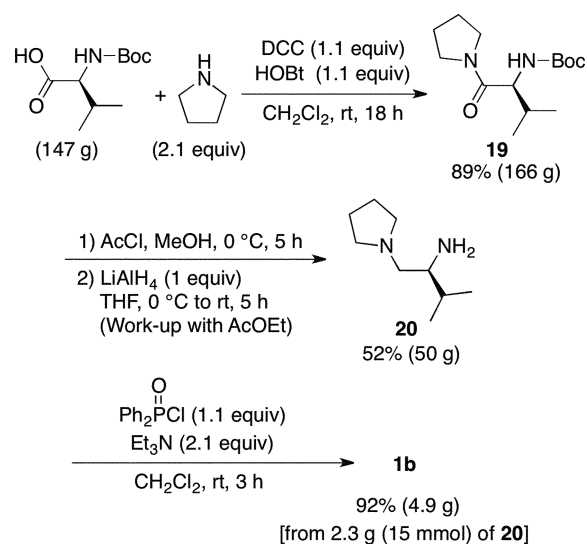
conditions for 3 h. According to the procedure developed by Zhao and co-workers,^{29a,b} the amount of Et₂Zn (3.0–4.5 equiv.; **15a**:Et₂Zn = 1:2 or 1:3) was examined. In the presence of chiral ligand **1a** (0.17 M; **9c**/toluene), compound **17a** was obtained in 70% yield with 94% ee when 3 equiv. of Et₂Zn was used (entry 1). However, self-aldol condensation was also observed, and **18** was obtained in 18% yield. In contrast, when 4.5 equiv. of Et₂Zn was used, the selectivity of the reactions was improved and **17a** was obtained in 86% yield with 94% ee (entry 2). The same enantioselectivity (94% ee) and reduced yield (79%) of **17a** were observed when the reaction was conducted under higher concentration conditions (0.25 M; **9c**/toluene) (entry 3), while the yield of **17a** was significantly reduced when the reaction was conducted under lower concentration (0.14 M; **9c**/toluene) conditions (entry 4). Moreover, when readily available ligand **1c** (*vide infra*) was used, **17a** was obtained in 83% yield and 94% ee, which was similar to the result with ligand **1a** (entries 2 and 5).

Under the optimized reaction conditions, a variety of ketones and arylboronic acids (**15**) were explored (Table 6). Not only acetylarenes (see **17b–f**) but also cyclic ketones such as 1-indanone (see **17g**) and α -tetralone (see **17h** and **17i**) could be used, and the corresponding products were obtained in good to high yields with 84–98% ee. Less-reactive *p*-fluorophenylboronic acid was acceptable for this reaction (see **17c** and **17i**). With regard to the ability to tolerate heteroaromatic

Table 6 Enantioselective arylation to ketones with arylboronic acids

Reaction scheme		Product, yield, and enantioselectivity	
$\text{ArB(OH)}_2 \xrightarrow[\text{toluene, MS 4Å, reflux, 3 h}]{\text{15 (1.5 equiv)}} \text{[ArBO]}_3 \xrightarrow[\text{toluene, 60 °C, 12 h}]{\text{Et}_2\text{Zn (4.5 equiv)}} \xrightarrow[\text{toluene, 0.17 M, rt, 24 h}]{\text{1a (10 mol%), R}^1\text{C(=O)R}^2 \text{ (9) (1 equiv)}} \text{HO-}^*\text{Ar-R}^1\text{R}^2 \text{ (17)}$			
		17b, 79%, 92% ee	
		17c, 72%, 92% ee	
		17d, 86%, 97% ee	
		17e, 82%, 92% ee	
		17f, 80%, 90% ee	
		17g, 85%, 84% ee	
		17h, 80%, 98% ee	
		17i, 52%, 96% ee	
		17j, 67%, 93% ee	
		17k, 76%, 90% ee	
		17l, 78%, 90% ee	
		17m, 71%, 77% ee	
		17n, 82%, 80% ee	

groups, the thienyl moiety could be used in a substrate or a reactant, and **17j** and **17k** were obtained in moderate to good yields in high enantioselectivities (90–93% ee). The 1,3-benzodioxolyl moiety as an oxyheterocycle was also used in a reactant (see **17l**). Furthermore, α,β -unsaturated ketones such as 1-acetyl-1-cyclohexene and benzylideneacetone were examined, and the

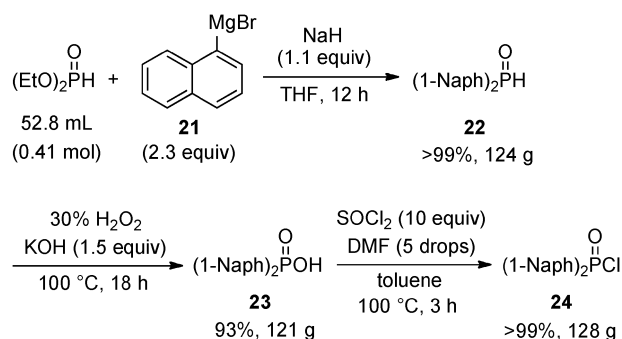
**Scheme 2** Preparation of chiral phosphoramidate ligand **1b**.

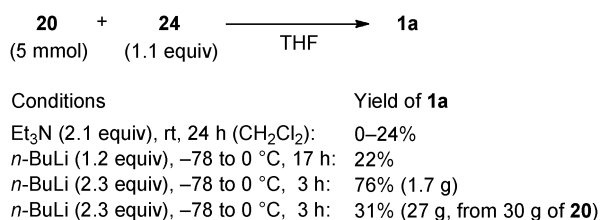
1,2-adducts (**17m** and **17n**) were obtained selectively as major products in good enantioselectivities (77–80% ee).

2.4. Preparation of L-valine-derived chiral phosphoramidate ligands

Chiral phosphoramidate ligand **1b** was readily prepared in multigram-scale quantities from commercially available *N*-Boc-L-valine in four steps, as shown in Scheme 2. First, the dehydrative amidation of *N*-Boc-L-valine (147 g, 0.677 mol) with pyrrolidine by treatment with *N,N'*-dicyclohexylcarbodiimide (DCC) and 1-hydroxybenzotriazole (HOBT) provided **19** in 89% yield (166 g). After deprotection of the Boc group by AcCl in methanol, diamine **20** was obtained in 52% yield (50 g) by reduction using LiAlH₄. Remarkably, one equivalent of LiAlH₄ was sufficient, and the reaction proceeded smoothly at 0 °C to room temperature within 5 h. Moreover, the use of ethyl acetate could control a safety work-up procedure without the generation of gaseous hydrogen even in a multigram-scale synthesis. Finally, phosphoramidation of **20** with commercially available diphenylphosphinic chloride was conducted in the presence of triethylamine, and **1b** was obtained in 92% yield (4.9 g) in a 15 mmol-scale synthesis.

On the other hand, for the synthesis of ligand **1a**, di(1-naphthyl)phosphinic chloride (**24**) was prepared on a >100 g-scale as shown in Scheme 3. Compound **24** was

**Scheme 3** Preparation of di(1-naphthyl)phosphinic chloride.

Scheme 4 Preparation of chiral phosphoramidate ligand **1a**.

obtained almost quantitatively in three steps, such as the Grignard addition reaction of diethyl phosphite with **21**, followed by oxidation of **22** with H₂O₂, and chlorination of **23** with SOCl₂.

However, in the phosphoramidation of **20** (5 mmol scale) with sterically bulky **24**, the reaction in the presence of triethylamine proceeded sluggishly, and **1a** was not obtained in reproducible yields in several examinations (0–24%) (Scheme 4). A lithium amide reagent prepared *in situ* from **20** and 1.2 equivalents of *n*-BuLi also showed low reactivity. In sharp contrast, the use of 2.3 equivalents of *n*-BuLi significantly improved the reactivity, and **1a** was obtained in 76% yield (1.7 g). In this ligand synthesis, dianion species may be involved and/or the corresponding acidic phosphoramidate N–H may be deprotonated by the remaining starting materials. Although it was difficult to completely control the strong exothermic reaction with **24** in a multigram-scale synthesis from **20** (30 g, 191 mmol), **1a** was obtained in 31% yield (27 g).

With the optimized method with 2.3 equiv. of *n*-BuLi or *t*-BuLi to **20**, some promising chiral phosphoramidate ligands were prepared on a 4–6 mmol scale, and the yields in phosphoramidation as a key step are shown in Fig. 1. A variety of sterically bulky chiral phosphoramidate ligands were synthesized in yields of 38–83%. Exceptionally, **1c** was scarcely detected (<10% yield) in the resulting complex mixture when 1.2 or 2.3 equiv. of *n*-BuLi was used, since

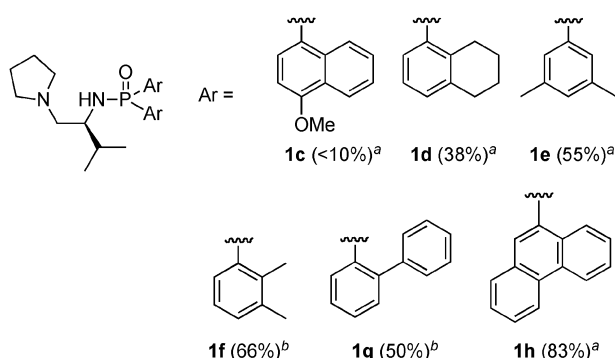
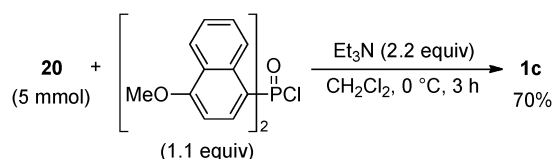
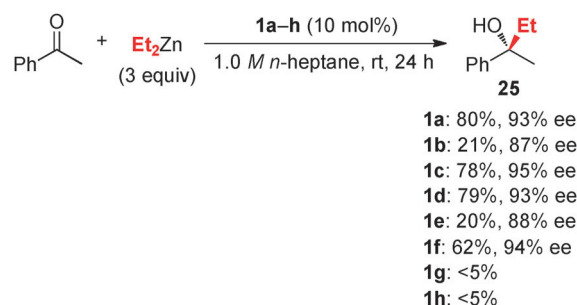


Fig. 1 Preparation of chiral phosphoramidate ligands with a dianion method. ^aYields in the phosphoramidation step with *n*-BuLi (2.3 equiv.). ^bYields in the phosphoramidation step with *t*-BuLi (2.3 equiv.).

Scheme 5 Preparation of chiral phosphoramidate ligand **1c**.

Scheme 6 Evaluation of chiral phosphoramidate ligands.

bis(4-methoxynaphthalen-1-yl)phosphinic chloride was probably too reactive under these conditions.

Fortunately, however, **1c** was synthesized again by using triethylamine (2.2 equiv.) in 70% yield, as shown in Scheme 5. Since ligand **1c**, like ligand **1a**, is valuable (*vide supra*), a convenient synthesis under mild reaction conditions such as triethylamine in place of *n*-BuLi is highly important, particularly when bulk quantities are needed for industrial use.

Chiral ligands **1a–h** were briefly evaluated in enantioselective diethylzinc addition to acetophenone as a probe reaction, as shown in Scheme 6. As well as ligands **1a** and **1c**, ligand **1d**, which contains tetrahydronaphthyl moieties, was effective. Ligands **1b** and **1e–g**, which include substituted-phenyl moieties, and ligand **1h**, which includes 9-phenanthryl moieties, showed low reactivity and/or enantioselectivity.

3. Conclusion

In summary, we have developed a highly practical technology, catalytic enantioselective alkyl and aryl addition to aldehydes and ketones with organozinc reagents, which were prepared *in situ* from Grignard reagents or arylboronic acids. Grignard reagents and boronic acids are highly attractive alkyl and aryl sources for both academic and industrial purposes, since a variety of these items are commercially available or readily prepared from the corresponding halides. In the Grignard reagent-derived di(2°-alkyl)zinc addition to aldehydes and ketones, refined Charette's reaction conditions with ZnCl₂/NaOMe/RMgCl in a molar ratio of 1/2.5/1.6 in the presence of our chiral phosphoramidate was essential. Optically active novel 2°-alcohols with two similar cyclic and/or acyclic fragments could be successfully synthesized and transformed to synthetically useful γ-hydroxy-β-pyrone, α-alkoxy carboxylic acid, α-hydroxy ketone, and 2,3-epoxyalcohol. Moreover, we achieved a concise synthesis of (*S*)-(+)-ginnol by taking advantage of a catalytic enantioselective long *n*-alkyl chain addition to aldehydes with Grignard reagent-derived dialkylzincs. Furthermore, we developed an arylzinc addition to ketones with arylboronic acids as an aryl source. In addition to arylboronic acids, heterocyclic boronic acids could also be used, and the corresponding optically active 3°-biaryl carbinols were obtained in good to high enantioselectivities. Chiral phosphoramidate ligands in this report were readily synthesized on a 1–30 gram scale, and the key to the development of unprecedented 2°-alkylation to aldehydes and arylation to less-reactive ketones is likely the high reactivity of these chiral ligands.

4. Experimental section

4.1. General procedure for the preparation of di(2°-alkyl)zinc reagents (Tables 1 and 2)

To a test tube equipped with a magnetic stirrer charged with ZnCl₂ (682 mg, 5 mmol) and NaOMe (676 mg, 12.5 mmol) was added Et₂O (5 mL) at room temperature under a nitrogen atmosphere. The suspension was stirred for 20 min and cooled to 0 °C for another 10 min. RMgCl in 2.0 M Et₂O solution (titrated, 4 mL, 8 mmol) was added dropwise with vigorous stirring over 10 min at 0 °C, and the suspension was allowed to stir at room temperature for 2 h. The mixture was centrifuged for 10 min (4000 rpm) and the di(2°-alkyl)zinc reagent (0.44 M in Et₂O) was gently transferred *via* a cannula into a well-dried Pyrex Schlenk tube to be stored before use. **Caution!** Centrifugation of pyrophoric organometallic reagents in high spinning should be done with a safety electromechanical centrifugal separator, particularly when large quantities are prepared.

4.2. General procedure for the catalytic enantioselective addition of di(2°-alkyl)zinc reagents to aldehydes (Tables 1 and 2)

A well-dried Pyrex Schlenk tube was charged with **1a** (22.8 mg, 0.05 mmol) and the di(2°-alkyl)zinc reagent (0.44 M in Et₂O) (3.4 mL, 1.5 mmol) at room temperature under a nitrogen atmosphere. Et₂O was removed under reduced pressure to generate the solvent free di(2°-alkyl)zinc reagent containing **1a in situ**. After the removal of Et₂O, toluene (0.4 mL) was added when (*c*-Hex)₂Zn was used. Aldehyde **2** (0.5 mmol) was added to the mixture at room temperature. The resulting mixture was stirred at room temperature for 2 h. Ether or EtOAc (10 mL) and sat. NH₄Cl aqueous solution (10 mL) were poured into the mixture at 0 °C. The product was extracted with ether (10 mL × 3) and washed with brine (10 mL). The combined extracts were dried over MgSO₄. The organic phase was concentrated under reduced pressure and the crude product was purified by neutral silica gel column chromatography (eluent: *n*-hexane/Et₂O or *n*-pentane/Et₂O) to give the desired products (**3**). The enantiomeric purity was determined by GC or HPLC on a chiral column.

4.3. General procedure for the catalytic enantioselective addition of di(2°-alkyl)zinc reagents to ketones (Table 3)

A well-dried Pyrex Schlenk tube was charged with **1a** (22.8 mg, 0.05 mmol) and the di(2°-alkyl)zinc reagent (0.44 M in Et₂O) (3.4 mL, 1.5 mmol) at room temperature under a nitrogen atmosphere. Et₂O was removed under reduced pressure to generate the solvent free di(2°-alkyl)zinc reagent containing **1a in situ**. After the removal of Et₂O, toluene (0.4 mL) was added when (*c*-Hex)₂Zn was used, and the mixture was cooled to 0 °C. Ketone (**9**) (0.5 mmol) was added to the mixture (in the case of **9b**, it was added dropwise at 0 °C over 12 h). The resulting mixture was stirred at 0 °C for 24 h. Ether or EtOAc (10 mL) and sat. NH₄Cl aqueous solution (10 mL) were poured into the mixture at 0 °C. The product was extracted with ether (10 mL × 3) and washed with brine (10 mL). The combined extracts were dried over MgSO₄. The organic phase was concentrated under reduced pressure and the crude product was purified by neutral silica gel column chromatography

(eluent: *n*-hexane/Et₂O = 20/1–8/1) to give the desired product (**10**). To determine the enantioselectivity by GC or HPLC on a chiral column, compound **10** was readily transformed to the corresponding acetate derivative by using Ac₂O/Et₃N/DMAP in CH₂Cl₂.

4.4. General procedure for the catalytic enantioselective long *n*-alkyl chain addition to aldehydes (Table 4)

A well-dried Pyrex Schlenk tube was charged with **1a** (22.8 mg, 0.05 mmol) and the R₂Zn reagent (prepared from ZnCl₂, NaOMe, and RMgCl as above, 0.4–0.6 M in Et₂O) (1.5 mmol) at room temperature under a nitrogen atmosphere. Et₂O was removed under reduced pressure to generate the solvent free R₂Zn reagent containing **1a in situ**. Aldehyde (0.5 mmol) was added to the mixture at room temperature. The resulting mixture was stirred at room temperature for 2 h. Ether (or *n*-hexane, toluene, EtOAc, *etc.*) (10 mL) and sat. NH₄Cl aqueous solution (10 mL) were poured into the mixture at 0 °C. The product was extracted with ether (10 mL × 3) and washed with brine (10 mL). The combined extracts were dried over MgSO₄. The organic phase was concentrated under reduced pressure and the crude product was purified by neutral silica gel column chromatography (eluent: *n*-hexane/Et₂O) to give the desired products (**12**). The enantiomeric purity was determined by GC on a chiral column.

4.5. Catalytic enantioselective synthesis of (*S*)-(+)–ginnol (**13**) (Scheme 1)

A well-dried Pyrex Schlenk tube was charged with **14** (68.7 mg, 0.10 mmol) and the (*n*-C₉H₁₉)₂Zn reagent (0.44 M in Et₂O) (3.4 mL, 1.5 mmol) at room temperature under a nitrogen atmosphere. Et₂O was removed under reduced pressure to generate the solvent free (*n*-C₉H₁₉)₂Zn reagent containing **14 in situ**. Toluene (1.5 mL) and THF (0.7 mL) were then added, and the mixture was stirred at room temperature for 1 h. Icosanal (148.3 mg, 0.5 mmol) was added. The resulting mixture was stirred at room temperature for 12 h. After hydrolysis with 10 mL of saturated NH₄Cl aqueous solution, the product was extracted with ethyl acetate (10 mL × 3) and washed with brine (10 mL). The combined extracts were dried over MgSO₄. The organic phase was concentrated under reduced pressure and the crude product was purified by neutral silica gel column chromatography (eluent: *n*-hexane/Et₂O) to give ginnol (172.5 mg, 81% yield). Enantioselectivity was confirmed by HPLC analysis of the diastereotopic (*R*)-MTPA-esters of the resulting ginnol. Chiral HPLC, Daicel chiralpack AD-3 × 2 at 4 °C, *n*-hexane/*i*-PrOH = 2000/1, flow rate = 0.1 mL min^{−1}, *t*_R = 100.3 min [major, (*S*)-derivative], 103.5 min [minor, (*R*)-derivative].

4.6. General procedure for the catalytic enantioselective arylzinc addition to ketones with arylboronic acids as an aryl source (Tables 5 and 6)

A solution of arylboronic acid (1.5 mmol) in toluene (3 mL) in a Pyrex Schlenk tube was stirred at room temperature for 10 min. The solution was heated under azeotropic reflux conditions with the removal of water. Water was removed through a pressure-equalized addition funnel that contained a

cotton plug and 4 Å molecular sieves (pellets) and functioned as a Soxhlet extractor. After heating for 3 h at reflux temperature (bath temp.; 140 °C), the reaction mixture was allowed to cool to ambient temperature. After the removal of volatiles under reduced pressure (<5 Torr), the corresponding triarylboroxine (**16**) was obtained *in situ* and could be used without purification. Et₂Zn (1.1 M in toluene) (4.1 mL, 4.5 mmol) and toluene (2 mL) were added at room temperature under a nitrogen atmosphere. The mixture was stirred at 60 °C for 12 h. Ligand **1a** (45.6 mg, 0.10 mmol) was added to the mixture and the solution was stirred for 10 min. Ketone **9** (1.0 mmol) was then added. The resulting mixture was stirred at room temperature for 24 h. After hydrolysis with 10 mL of sat. NH₄Cl aqueous solution, the product was extracted with ether (10 mL × 3) and washed with brine (10 mL). The combined extracts were dried over MgSO₄. The organic phase was concentrated under reduced pressure and the crude product was purified by neutral silica gel column chromatography (eluent: *n*-hexane/Et₂O) to give the desired products (**17**). The enantiomeric purity was determined by HPLC on a chiral column.

Acknowledgements

Financial support was provided by JSPS. KAKENHI (20245022), MEXT. KAKENHI (21750094, 21200033), and the Global COE Program of MEXT. We are grateful to Tosoh Finechem Corp. for providing organometallic reagents.

Notes and references

- For reviews: (a) K. Soai and S. Niwa, *Chem. Rev.*, 1992, **92**, 833; (b) L. Pu and H.-B. Yu, *Chem. Rev.*, 2001, **101**, 757; (c) M. Hatano, T. Miyamoto and K. Ishihara, *Curr. Org. Chem.*, 2007, **11**, 127; (d) M. Hatano and K. Ishihara, *Synthesis*, 2008, 1647.
- (a) K. Soai, T. Hayase, K. Takai and T. Sugiyama, *J. Org. Chem.*, 1994, **59**, 7908; (b) K. Soai, T. Shibata, H. Morioka and K. Choji, *Nature*, 1995, **378**, 767; (c) T. Hayase, T. Sugiyama, M. Suzuki, T. Shibata and K. Soai, *J. Fluorine Chem.*, 1997, **84**, 1; (d) T. Shibata, H. Tabira and K. Soai, *J. Chem. Soc., Perkin Trans. 1*, 1998, 177; (e) M. Asami, H. Watanabe, K. Honda and S. Inoue, *Tetrahedron: Asymmetry*, 1998, **9**, 4165; (f) W. K. Yang and B. T. Cho, *Tetrahedron: Asymmetry*, 2000, **11**, 2947; (g) I. Sato, R. Kodaka, K. Hosoi and K. Soai, *Tetrahedron: Asymmetry*, 2002, **13**, 805.
- Textbooks for the preparation of dialkylzinc reagents: (a) M. Schlosser, *Organometallics in Synthesis: A Manual*, Wiley, Chichester, 2nd edn, 2001; (b) H. Yamamoto and K. Oshima, *Main Group Metals in Organic Synthesis*, Wiley-VCH, Weinheim, 2004; (c) P. Knochel, *Handbook of Functionalized Organometallics*, Wiley-VCH, Weinheim, 2005.
- Recent selected papers for the preparation of dialkylzinc reagents: (a) P. Wipf and W. Xu, *Tetrahedron Lett.*, 1994, **35**, 5197; (b) B. H. Lipshutz, M. R. Wood and R. Tirado, *J. Am. Chem. Soc.*, 1995, **117**, 6126; (c) S. Berger, F. Langer, C. Lutz, P. Knochel, T. A. Mobley and C. K. Reddy, *Angew. Chem., Int. Ed. Engl.*, 1997, **36**, 1496; (d) C. Lutz, P. Jones and P. Knochel, *Synthesis*, 1999, 312; (e) C. Bolm, N. Hermanns, J. P. Hildebrand and K. Muñiz, *Angew. Chem., Int. Ed.*, 2000, **39**, 3465; (f) S. Dahmen and S. Bräse, *Org. Lett.*, 2001, **3**, 4119; (g) A. Rimkus and N. Sewald, *Org. Lett.*, 2002, **4**, 3289; (h) S.-J. Jeon, H. Li and P. J. Walsh, *J. Am. Chem. Soc.*, 2005, **127**, 16416; (i) J. G. Kim and P. J. Walsh, *Angew. Chem., Int. Ed.*, 2006, **45**, 4175; (j) L. Salvi, J. G. Kim and P. J. Walsh, *J. Am. Chem. Soc.*, 2009, **131**, 12483.
- Harada and co-workers reported a catalytic enantioselective 1°-alkylation and arylation to aldehydes with Grignard reagent/Ti(Oi-Pr)₄. (a) Y. Muramatsu and T. Harada, *Angew. Chem., Int. Ed.*, 2008, **47**, 1088; (b) Y. Muramatsu and T. Harada, *Chem.-Eur. J.*, 2008, **14**, 10560; (c) Y. Muramatsu, S. Kanehira, M. Tanigawa, Y. Miyawaki and T. Harada, *Bull. Chem. Soc. Jpn.*, 2010, **83**, 19. Recently, they reported a catalytic enantioselective synthesis of allylic alcohols from aldehydes and alkenylboron reagents. (d) T. Shono and T. Harada, *Org. Lett.*, 2010, **12**, 5270.
- A. Côté and A. B. Charette, *J. Am. Chem. Soc.*, 2008, **130**, 2771.
- (a) M. Hatano, S. Suzuki and K. Ishihara, *J. Am. Chem. Soc.*, 2006, **128**, 9998; (b) M. Hatano, S. Suzuki and K. Ishihara, *Synlett*, 2010, 321.
- P. J. Walsh, H. Li and C. A. de Parrodi, *Chem. Rev.*, 2007, **107**, 2503.
- H. Soroos and M. Morgana, *J. Am. Chem. Soc.*, 1944, **66**, 893.
- (a) M. Hatano, T. Miyamoto and K. Ishihara, *Org. Lett.*, 2007, **9**, 4535; (b) M. Hatano, T. Mizuno and K. Ishihara, *Synlett*, 2010, 2024; (c) M. Hatano, T. Mizuno and K. Ishihara, *Chem. Commun.*, 2010, **46**, 5443; (d) M. Hatano, T. Mizuno and K. Ishihara, *Tetrahedron*, 2011, **67**, 4417.
- MIB is an advantageous alternative to Noyori's DAIB [3-*exo*-(dimethylamino)isoborneol]. (a) M. Kitamura, S. Suga, K. Kawai and R. Noyori, *J. Am. Chem. Soc.*, 1986, **108**, 6071; (b) R. Noyori, S. Suga, K. Kawai, S. Okada, M. Kitamura, N. Oguni, M. Hayashi, T. Kaneko and Y. Matsuda, *J. Organomet. Chem.*, 1990, **382**, 19; (c) W. A. Nugent, *Chem. Commun.*, 1999, 1369; (d) R. Rosner, P. J. Sears, W. A. Nugent and D. G. Blackmond, *Org. Lett.*, 2000, **2**, 2511.
- Reviews and perspectives for the catalytic asymmetric hydrogenation and transfer hydrogenation of ketones: (a) R. Noyori and S. Hashiguchi, *Acc. Chem. Res.*, 1997, **30**, 97; (b) W. S. Knowles, *Angew. Chem., Int. Ed.*, 2002, **41**, 1998; (c) R. Noyori, *Angew. Chem., Int. Ed.*, 2002, **41**, 2008; (d) R. Noyori, M. Kitamura and T. Ohkuma, *Proc. Natl. Acad. Sci. U. S. A.*, 2004, **101**, 5356; (e) W. S. Knowles and R. Noyori, *Acc. Chem. Res.*, 2007, **40**, 1238.
- K. Cheng, A. R. Kelly, R. A. Kohn, J. F. Dweck and P. J. Walsh, *Org. Lett.*, 2009, **11**, 2703.
- N. L. Holder, *Chem. Rev.*, 1982, **82**, 287.
- M. Kusakabe, Y. Kitano, Y. Kobayashi and F. Sato, *J. Org. Chem.*, 1989, **54**, 2085.
- H. Gröger, *Adv. Synth. Catal.*, 2001, **343**, 547.
- K. M. Miller, W.-S. Huang and T. F. Jamison, *J. Am. Chem. Soc.*, 2003, **125**, 3442.
- (a) F. A. Davis and B. C. Chen, *Chem. Rev.*, 1992, **92**, 919; (b) L. A. Paquette, S. V. O'Neil, N. Guillo, Q. Zeng and D. G. Young, *Synlett*, 1999, 1857; (c) P. Hoyos, J.-V. Sinisterra, F. Molinari, A. R. Alcántara and P. D. de María, *Acc. Chem. Res.*, 2010, **43**, 288.
- (a) K. B. Sharpless and R. C. Michaelson, *J. Am. Chem. Soc.*, 1973, **95**, 6136; (b) C. M. Marson, A. J. Walker, J. Pickering, S. Harper, R. Wigglesworth and S. J. Edge, *Tetrahedron*, 1993, **49**, 10317; (c) W. Adam and T. Wirth, *Acc. Chem. Res.*, 1999, **32**, 703; (d) M. H. Hussain and P. J. Walsh, *Acc. Chem. Res.*, 2008, **41**, 883.
- A. H. Hoveyda, D. A. Evans and G. C. Fu, *Chem. Rev.*, 1993, **93**, 1307.
- Recent asymmetric synthesis of tertiary alcohols with 2°-nucleophiles. (a) K. Oisaki, D. Zhao, M. Kanai and M. Shibasaki, *J. Am. Chem. Soc.*, 2006, **128**, 7164; (b) J. L. Stymiest, V. Bagutski, R. M. French and V. K. Aggarwal, *Nature*, 2008, **456**, 778; (c) F. Wang, X. Liu, Y. Zhang, L. Lin and X. Feng, *Chem. Commun.*, 2009, 7297.
- (a) F. Langer, J. Waas and P. Knochel, *Tetrahedron Lett.*, 1993, **34**, 5261; (b) C. Eisenberg and P. Knochel, *J. Org. Chem.*, 1994, **59**, 3760; (c) R. Ostwald, P.-Y. Chavant, H. Stadtmueller and P. Knochel, *J. Org. Chem.*, 1994, **59**, 4143; (d) S. Vettel, A. Vaupel and P. Knochel, *Tetrahedron Lett.*, 1995, **36**, 1023; (e) S. Vettel, A. Vaupel and P. Knochel, *J. Org. Chem.*, 1996, **61**, 7173; (f) C. Lutz, C.-D. Graf and P. Knochel, *Tetrahedron*, 1998, **54**, 10317; (g) A. Fürstner, J. Mlynarski and M. Albert, *J. Am. Chem. Soc.*, 2002, **124**, 10274; (h) A. Fürstner, M. Albert, J. Mlynarski, M. Matheu and E. DeClercq, *J. Am. Chem. Soc.*, 2003, **125**, 13132; (i) S.-J. Jeon, H. Li, C. García, L. K. LaRochelle and P. J. Walsh, *J. Org. Chem.*, 2005, **70**, 448.
- It has been reported that LiCl generally improves the activity of Grignard reagents. (a) A. Krasovskiy and P. Knochel, *Angew. Chem., Int. Ed.*, 2004, **43**, 3333; (b) D. R. Armstrong, P. García-Alvarez, A. R. Kennedy, R. E. Mulvey and J. A. Parkinson, *Angew. Chem., Int. Ed.*, 2010, **49**, 3185.

- 24 (a) S. Beckmann and H. Schühle, *Z. Naturforsch., B: Anorg. Chem. Org. Chem. Biochem. Biophys. Biol.*, 1968, **23**, 471; (b) J.-H. Fuhrhop, T. Bedurke, A. Hahn, S. Grund, J. Gatzmann and M. Riederer, *Angew. Chem., Int. Ed. Engl.*, 1994, **33**, 350; (c) T. Kusumi, H. Takahashi, T. Hashimoto, Y. Kan and Y. Asakawa, *Chem. Lett.*, 1994, 1093; (d) L. Schwink and P. Knochel, *Tetrahedron Lett.*, 1994, **35**, 9007; (e) F. Langer, L. Schwink, A. Devasagayraj, P.-Y. Chavant and P. Knochel, *J. Org. Chem.*, 1996, **61**, 8229; (f) T. Berkenbusch and R. Brückner, *Tetrahedron*, 1998, **54**, 11471; (g) A. Dommissie, J. Wirtz, K. Koch, W. Barthlott and T. Kolter, *Eur. J. Org. Chem.*, 2007, 3508.
- 25 (a) M. Hatano, T. Miyamoto and K. Ishihara, *Adv. Synth. Catal.*, 2005, **347**, 1561; (b) M. Hatano, T. Miyamoto and K. Ishihara, *Synlett*, 2006, 1762; (c) M. Hatano, T. Miyamoto and K. Ishihara, *J. Org. Chem.*, 2006, **71**, 6474.
- 26 Pioneering report of the catalytic enantioselective addition of Ph_2Zn to ketones. (a) P. I. Dosa and G. C. Fu, *J. Am. Chem. Soc.*, 1998, **120**, 445; (b) C. García and P. J. Walsh, *Org. Lett.*, 2003, **5**, 3641; (c) H. Li, C. García and P. J. Walsh, *Proc. Natl. Acad. Sci. U. S. A.*, 2004, **101**, 5425.
- 27 For a review of catalytic enantioselective arylation. (a) C. Bolm, J. P. Hildebrand, K. Muñoz and N. Hermanns, *Angew. Chem., Int. Ed.*, 2001, **40**, 3284; (b) F. Schmidt, R. T. Stemmler, J. Rudolph and C. Bolm, *Chem. Soc. Rev.*, 2006, **35**, 454; (c) M. W. Paixão, A. L. Braga and D. S. Lüdtke, *J. Braz. Chem. Soc.*, 2008, **19**, 813.
- 28 (a) C. Bolm and J. Rudolph, *J. Am. Chem. Soc.*, 2002, **124**, 14850; (b) J. Rudolph, F. Schmidt and C. Bolm, *Synthesis*, 2005, 840.
- 29 (a) C. Jimeno, S. Sayalero, T. Fjermestad, G. Colet, F. Maseras and M. A. Pericàs, *Angew. Chem., Int. Ed.*, 2008, **47**, 1098; (b) X. Wu, X. Liu and G. Zhao, *Tetrahedron: Asymmetry*, 2005, **16**, 2299; (c) Z. Chai, X.-Y. Liu, X.-Y. Wu and G. Zhao, *Tetrahedron: Asymmetry*, 2006, **17**, 2442; (d) C. Liu, Z.-L. Guo, J. Weng, G. Lu and A. S. C. Chan, *Chirality*, 2010, **22**, 159.
- 30 Catalytic enantioselective arylation of ketones using arylboronic acids or triarylboroxines and Et_2Zn . (a) O. Prieto, D. J. Ramón and M. Yus, *Tetrahedron: Asymmetry*, 2003, **14**, 1955; (b) V. J. Forrat, O. Prieto, D. J. Ramón and M. Yus, *Chem.-Eur. J.*, 2006, **12**, 4431.
- 31 Catalytic enantioselective rhodium and palladium-catalyzed addition of arylboronic acids to ketones. (a) S. L. X. Martina, R. B. C. Jagt, J. G. de Vries, B. L. Feringa and A. J. Minnaard, *Chem. Commun.*, 2006, 4093; (b) G. Liu and X. Lu, *J. Am. Chem. Soc.*, 2006, **128**, 16504; (c) P. He, Y. Lu, C.-G. Dong and Q.-S. Hu, *Org. Lett.*, 2007, **9**, 343; (d) H.-F. Duan, J.-H. Xie, X.-C. Qiao, L.-X. Wang and Q.-L. Zhou, *Angew. Chem., Int. Ed.*, 2008, **47**, 4351.